

Application Development Specification for A&D Bluetooth[®] BLE Series Blood pressures monitor UA-651BLE

Bluetooth Low Energy

History

Version	Description	Date
0.8	Ver 1.0	02.Apr.14
0.9	Add Information to Profile format(1.1.7~1.1.9) Modification of Communications Sequence (After Confirmation)	15.May.14
1.0	Feature's endian changed Big to Little Add Device Information Service	07.Jul.14
1.1	Date Time, Battery Service Change the value, Separate each device.	04.Aug.14
1.2	Add to the explanation with supplement(2.1.1)	27.Oct.14

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1 General Information

This document describes the communication specification for A&D BLE devices.

A&D BLE communication is consist of two parts, using profiles which Bluetooth SIG has already released and using A&D proprietary profile.

2 Services in A&D devices

A&D BLE devices are implemented Standard profiles, including each device service, Battery Service, Device Information Service, except for Weight Scale, and proprietary A&D service which provides commands to set a device and to check device condition.

2.1 Blood Pressure Service (0x1810) for A&D device

2.1.1 Blood Pressure Measurement (0x2A35)

Name	Byte	Format	Information			
Flags	1	8bit	Bit	Name	Key	Value
			0	Blood Pressure Unit Flag	0	mmHg (C1)
					1	kPa (C2)
			1	Time Stamp Flag	0	Time Stamp not present
					1	Time Stamp present(C3)
			2	Pulse Rate Flag	0	Pulse Rate not present
					1	Pulse Rate present (C4) (fixed)
			3	User ID Flag	0	User ID not present (fixed)
					1	User ID present (C5)
			4	Measurement Status Flag	0	Measurement Status not present
					1	Measurement Status present(C6)(fixed)
Systolic Pressure(mmHg)(C1)	2	SFLOAT	Systolic Pressure(mmHg) (C1) ex) 128mmHg → 0x80, 0x00			
Diastolic Pressure(mmHg)(C1)	2	SFLOAT	Diastolic Pressure (mmHg) (C1)			
Mean Arterial Pressure(mmHg)(C1)	2	SFLOAT	Mean Arterial Pressure (mmHg) (C1)			
Systolic Pressure(kPa) (C2)	2	SFLOAT	Systolic Pressure (kPa) (C2) ex) 17.6Kpa → 0xB0, 0xF0			
Diastolic Pressure(kPa) (C2)	2	SFLOAT	Diastolic Pressure (kPa) (C2)			
Mean Arterial Pressure (kPa) (C2)	2	SFLOAT	Mean Arterial Pressure (kPa) (C2)			
Time Stamp (C3)	7	Date	Time Stamp ex) 2013/8/26 9:10:20→0xDD 0x07 0x08 0x1A 0x09 0x0A 0x14			
Pulse Rate (C4)	2	SFLOAT	Pulse Rate Ex) 60/min → 0x3C, 0x00			
User ID (C5)	1	uint8	Key		Value	
			255		Unknown User	
			0-254		Defined by service specification	
Measurement Status (C6)	2	flag	Bit	Name	Key	Value
			0	Body Movement Detection Flag	0	No body movement (fixed)
					1	Body movement during measurement
			1	Cuff Fit Detection Flag	0	11
					1	Cuff too loose
			2	Irregular Pulse Detection Flag	0	No irregular pulse detected
					1	Irregular pulse detected
			3-4	Pulse Rate Range Detection Flags	0	Pulse rate is within the range
					1	Pulse rate exceeds upper limit
					2	Pulse rate is less than lower limit
					3	Reserved for future use
			5	Measurement Position Detection Flag	0	Proper measurement position
					1	Improper measurement position
6-15		Reserved for future use				

- Measurement Error : C1,C2,C4 value is 0xFF, 0x07
- Pulse Rate Range Error : C4 value is 0x00, 0x08. Display "E" in LCD, BP can send upper limit and Lower limit detection.
- Not Time Settings : Flags Bit1=0 and C3 value is **N.A(refer to 3.2)**.
- "Measurement Status" is **Little Endian**
Irregular Pulse Detection: C6 value is 0x04, 0x00

2.1.2 Blood Pressure Feature (0x2A49)

Name	Byte	Format	Information			
Blood Pressure Feature	2	16bit	Bit	Name	Key	Value
			0	Body Movement Detection Support bit	0	Body Movement Detection feature not supported (fixed)
					1	Body Movement Detection feature supported
			1	Cuff Fit Detection Support bit	0	Cuff Fit Detection feature not supported (fixed)
					1	Cuff Fit Detection feature supported
			2	Irregular Pulse Detection Support bit	0	Irregular Pulse Detection feature not supported
					1	Irregular Pulse Detection feature supported (fixed)
			3	Pulse Rate Range Detection Support bit	0	Pulse Rate Range Detection feature not supported
					1	Pulse Rate Range Detection feature supported
			4	Measurement Position Detection Support bit	0	Measurement Position Detection feature not supported
					1	Measurement Position Detection feature supported
			5	Multiple Bond Support bit	0	Multiple Bonds not supported (fixed)
					1	Multiple Bonds supported
6-15	Reserved for future use					

- "Blood Pressure Feature" is **Little Endian**
Irregular Pulse Support: value is 0x0c, 0x00

2.1.3 Client Characteristic Configuration descriptor

Blood Pressure Measurement is sent with Indication

Name	Byte	Format	Information		
Flags	1	16bit	Bit	Key	Value
			0	0	Notifications disabled
				1	Notifications enabled
			1	0	Indications disabled
				1	Indications enabled
			2-15		Reserved for future use

- "CCCD" is **Little Endian**.

Indication 0x02, 0x00

2.1.4 Date and Time

Date & Time characteristic is used to set time and date to A&D device and to read time and date from A&D device.

Name	Byte	Format	Value
Year	2	16bit	07DD~08CF
Month	1	8bit	01~0C
Day	1	8bit	01~1F
Hours	1	8bit	00~18
Minutes	1	8bit	00~3B
Second	1	8bit	00~3B

One of the parts, year, month, date, hour, minute and second would be out of the range, the device cannot update the time and date.

2.2 Battery Service (0x180F) Common operation for each device.

2.2.1 Battery Level (0x2A19)

Name	Byte	Format	Level	Value
Level	1	uint8	100%	0x64
			66%	0x42
			40%	
			33%	0x21

Each device has its own battery level indicator and sends rate of battery level defined by each device. Devices can take a measurement when a device can send battery information.

2.3 Device information Service Common operation for each device.

Every device uses Device Information service including Model Number String, Firmware Revision String, Registration Certification Data, Serial Number String and System ID.

All string information have to apply with even number of characters.

2.3.1 Manufacture Name String (0x2A29)

Byte	Format	Value
12	Ascii	A&D Medical _

2.3.2 Model Number String (0x2A24)

Byte	Format	Value
10	Ascii	UA-651BLE_

2.3.3 Serial Number String (0x2A25)

Byte	Format	Value
10	Ascii	Ex)5140700001

2.3.4 Hardware Revision String (0x2A27)

Byte	Format	Value
4	Ascii	Ex)0.00

2.3.5 Firmware Revision String (0x2A26)

Byte	Format	Value
12	Ascii	Ex)BLP008_d015_

2.3.6 Software Revision String (0x2A28)

Byte	Format	Value
4	Ascii	Ex)0.00

2.3.7 System ID (0x2A23)

XXXXXXXX XXXXXX ; BD Address except for FFFE

Byte	Format	Value
8	uint8	xxxxxxFFFExxxxxx

2.3.8 Registration Certification Data (0x2A2A)

Byte	Format	Value
22	uint8	Ex)00020012020100080400000100028007020200020000 ※see next page

※_: Represent space

※Registration Certification Data

Byte	Description	data	value
2	RegCertDataList.count	2	0x00 0x02
2	RegCertDataList.length	18	0x00 0x12
-	RegCertDataList[0]		
1	auth-body	continua = 2	0x02
1	auth-body-struct-type	Body Struct = 1	0x01
2	auth-body-data.length	8	0x00 0x08
-	auth-body-data		
1	ContinuBodyStruct.major-IG-version	DG2014	0x04
1	ContinuBodyStruct.minor-IG-version	DG2014	0x00
2	CertifiedDeviceClassList.count	1	0x00 0x01
2	CertifiedDeviceClassList.length	2	0x00 0x02
-	CertifiedDeviceClassList[0]		
-	MDC_DEV_SPEC_PROFILE	4103:Blood Pressure	
2	CertifiedDeviceClass	Monitor_Upper arm (see. ISO11073-10101)	0x80 0x07
-	RegCertDataList[1]		
1	auth-body	continua = 2	0x02
1	auth-body-struct-type	RegStruct = 2	0x02
2	RegCertDataList[1]	2	0x00 0x02
2	This is a regulated device	0	0x00 0x00

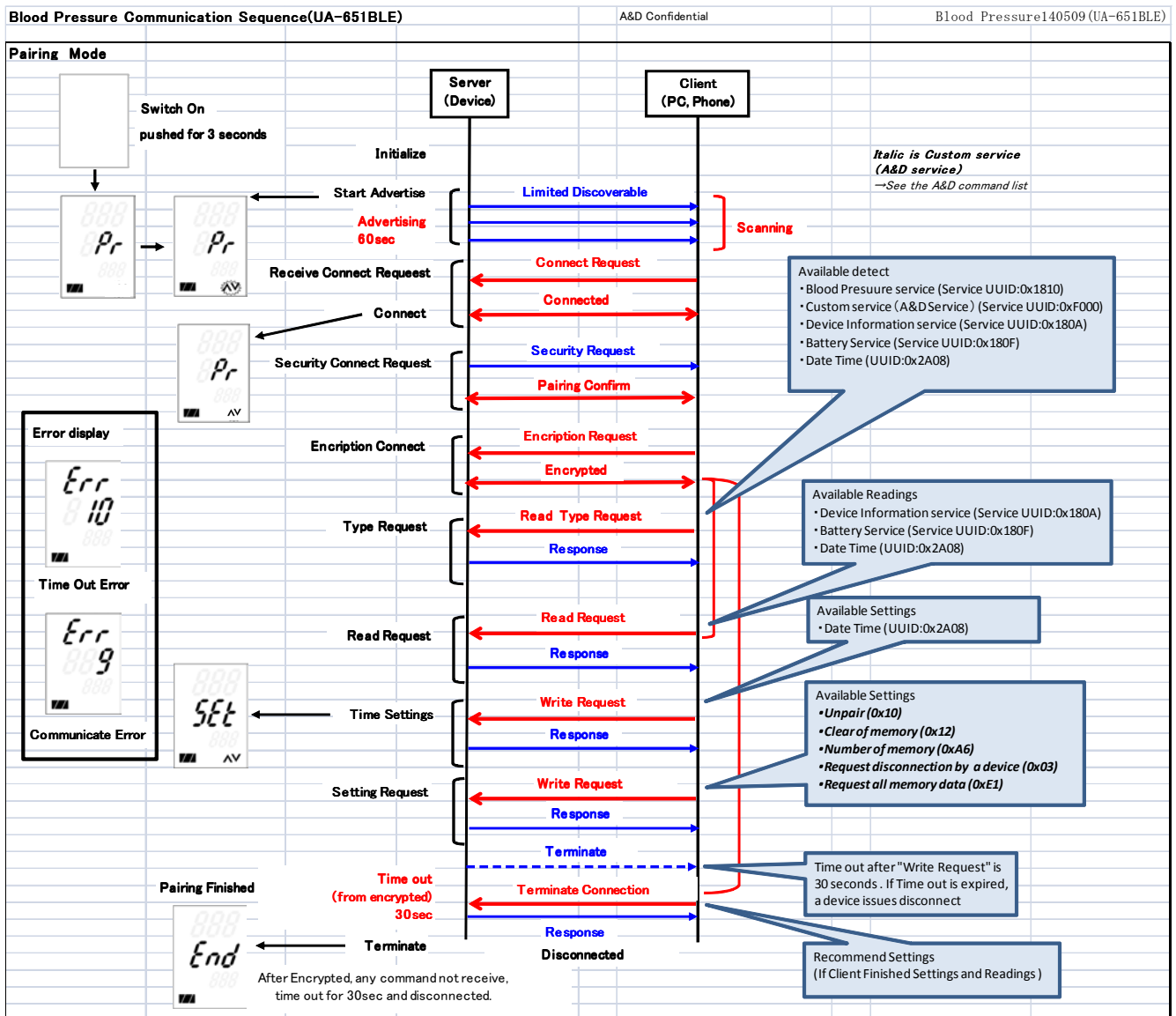
3 Pairing and Communications Sequence

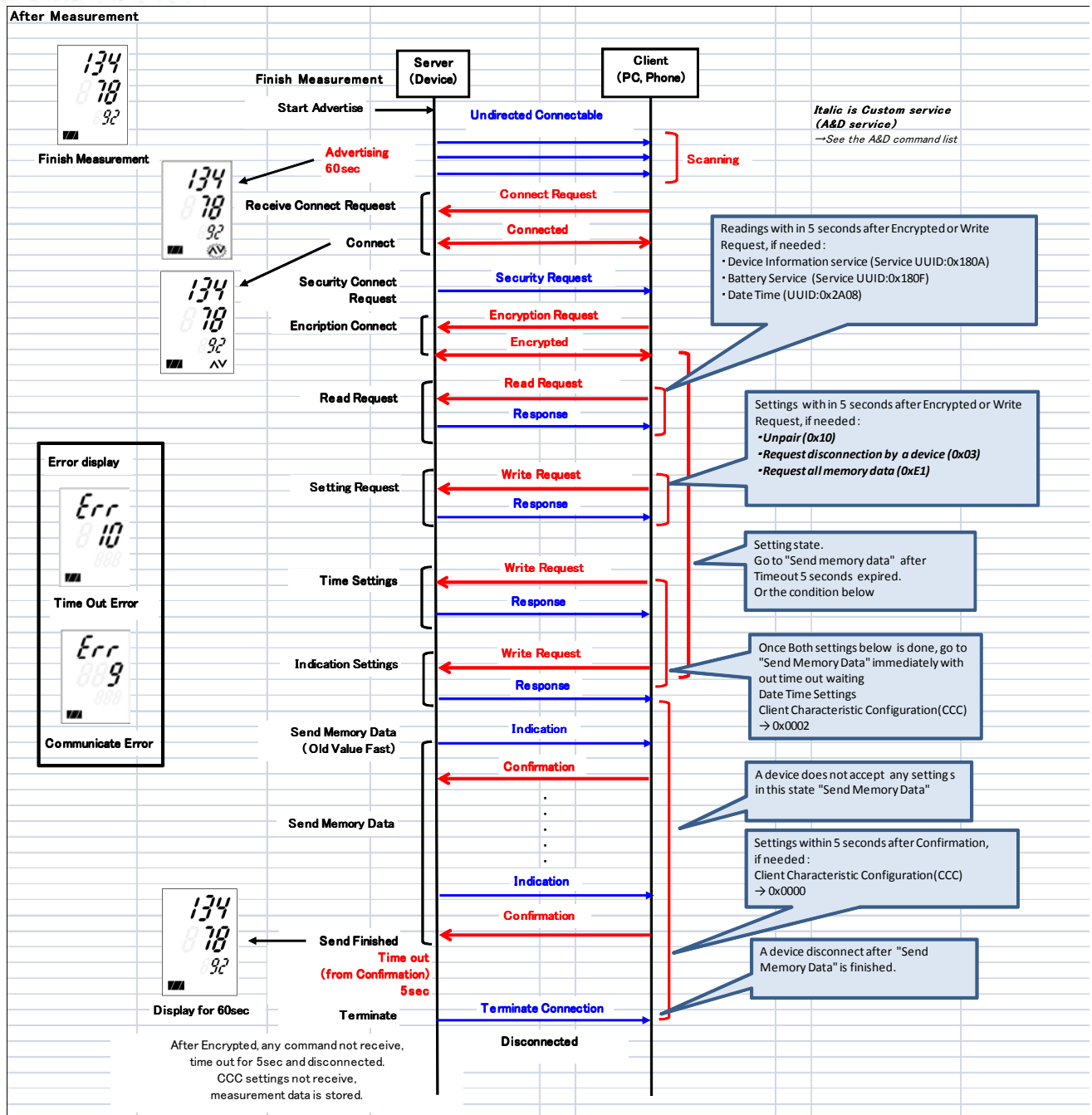
This section describes operation sequence of Pairing and Data Communication.

1. Set the application to pairing mode to start scanning.
2. Start pairing of A&D BLE device following each instruction manual.
3. At pairing mode, the application should set time and date and any other device settings to A&D BLE device. After successful pairing, A&D BLE device shows "End" on the screen.
4. Take a measurement and finish the measurement, then A&D BLE device start BLE connection with advertising. The application starts scanning with suitable interval so that the application catches the advertising of A&D BLE device as soon as it can.
5. At initial connection or pairing, the Application set "2" to CCCD (Client Characteristic Configuration Descriptor) so that A&D BLE device sends a measurement data with Indication.
6. After A&D device recognizes to be set "2" to CCCD and to be synchronized time and date within 5 seconds after connected, send the data with Indication.
7. If the timeout set CCCD and time and date is expired, A&D BLE device will not send data and store the data in memory. The stored data in A&D BLE device can send next successful connection.

Please refer GATT specification at Bluetooth Developer Portal to investigate Profile, Service and Characteristic in detail. Ref. Portal (<https://developer.bluetooth.org/Pages/default.aspx>)

3.1 Blood Pressure Monitor (UA-651BLE)





3.2 Time Setting (All Devices)

A&D BLE Device requires to synchronize time and date with the application at every connection and pairing. If A&D BLE device has not adjust time and date after battery change, the device send or store measurement data without time and date.

To check time set in the device, the application can read Date & Time (0x2A08) with a few minutes delay due to writing timing or can use command 0x04 in A&D Service.

Blood Pressure Monitor UA-651BLE does not support to read time using both methods.

4 Profile Mapping

4.1 Blood Pressure Monitor (UA-651BLE)

Blood Pressure Monitor

Handle	UUID Type	UUID	Attribute Type	Parameter Values or Description	Format	Endian	Permission	Len	Register	Value	Read/W rite	Comment
0x0001		0x2800	Primary Service	GAP Service (0x1800)								
0x0002		0x2803	Characteristic Definition									
0x0003		0x2A00	Device Name									
0x0004		0x2803	Characteristic Definition									
0x0005		0x2A01	Appearance									
0x0006		0x2803	Characteristic Definition	0x0000								
0x0007		0x2A02	Peripheral Privacy Flag	0x00								
0x0008		0x2803	Characteristic Definition									
0x0009		0x2A03	Reconnection Address	00:00:00:00:00:00								
0x000A		0x2803	Characteristic Definition									
0x000B		0x2A04	Peripheral Preferred Connection Interval	0x0050, 0x00A0, 0x0000, 0x0258								
0x000C		0x2800	Primary Service	GATT Service (0x1801)								
0x000D		0x2803	Characteristic Definition									
0x000E		0x2A05	Service Changed									
0x000F		0x2902	Client Characteristic Configuration	0x0000 (default)								
0x0010	16	0x2800	Primary Service	Blood Pressure Service(0x1810)	UInt16		1	17				
0x0011	16	0x2803	Characteristic Definition		UInt8		1	1	20			
0x0012	16	0x2A35	Blood Pressure Measurement	Blood Pressure Measurement Value Set by Host(20 octets max)	Ascii	Little	0	20	0x2310		Indicate	
0x0013	16	0x2902	Client Characteristic Configuration	CCC Descriptor for Weight Measurement Characteristic(2 octets)	UInt8		3	19	0x2313			
0x0014	16	0x2803	Characteristic Definition		UInt8		1	1	02			
0x0015	16	0x2A49	Blood Pressure Feature	Blood Pressure Feature Characteristic Value Set by Host(2 octet)	Ascii	Little	1	20	0x2320 0004		R	
0x0016	16	0x2803	Characteristic Definition		UInt8		1	1	0A			
0x0017	16	0x2A08	Date Time	Date Time Characteristic Value Set by Host(7octets)	Ascii		3	20	0x2340		R,W	

Confidential information

Application Development Specification for A&D Bluetooth BLE Series

Handle	UUID Type	UUID	Attribute Type	Parameter Values or Description	Format	Endian	Permission	Len	Register	Value	Read/Write	Comment
0x0018	16	0x2800	Primary Service	Device Information Service (0x180A)	UInt16		1	3		180A		
0x0019	16	0x2803	Characteristic Definition		UInt8		1	1		02		
0x001A	16	0x2A29	Manufacture Name String		Ascii	Big	1	11		A&D Medical	R	Company Name
0x001B	16	0x2803	Characteristic Definition		UInt8		1	1		02		
0x001C	16	0x2A24	Model Number String		Ascii	Big	1	9		UA-651BLE	R	Model Name
0x001D	16	0x2803	Characteristic Definition		UInt8		1	1		02		
0x001E	16	0x2A25	Serial Number String	Serial Number String Characteristic Value Set by Host(20octets max)	Ascii	Big	1	10	0x2360	5140100021(Ex)	R	Device Serial Number
0x001F	16	0x2803	Characteristic Definition		UInt8		1	1		02		
0x0020	16	0x2A27	Hardware Revision String		Ascii	Big	3	4		0.00	R	Module Board Version
0x0021	16	0x2803	Characteristic Definition		UInt8		1	1		02		
0x0022	16	0x2A26	Firmware Revision String		Ascii	Big	1	6		BLP008.d013	R	Profile, Device Version
0x0023	16	0x2803	Characteristic Definition		UInt8		1	1		02		
0x0024	16	0x2A28	Software Revision String		Ascii	Big	1	4		0.00	R	BLE Stack Version
0x0025	16	0x2803	Characteristic Definition		UInt8		1	1		0A		
0x0026	16	0x2A23	System ID	System ID Characteristic Value Set by Host(20octets max)	UInt8	Little	3	16	0x2370	xxxxxxFFExxxxx	R	xx BT Address
0x0027	16	0x2803	Characteristic Definition		UInt8		1	1		0A		
0x0028	16	0x2A2A	Registration Certification Data		UInt8	Big	3	44		000200120201000 804000001000280 07020200020000	R	See "RegCertData"
0x0029	16	0x2800	Primary Service	Battery Service (0x180F)	UInt16		1	3		180F		
0x002A	16	0x2803	Characteristic Definition		UInt8		1	1		02		
0x002B	16	0x2A19	Battery Level	Battery Level Characteristic Value Set by Host(1 octet)	Ascii		1	20	0x2350		R	
0x002C	16	0x2800	Primary Service Custom Service	Custom Service(0xF000)	UInt16		1	17		233BF0005A341B 6D975C000D5690 ABE4		
0x002D	16	0x2803	Characteristic Definition		UInt8		1	1		0A		
0x002E	128	233BF0015A3 41B6D975C00 0D5690ABE4	Custom Characteristic	Custom Characteristic Value Set by Host(20octets max)	Ascii	Big	3	20	0x2330		R,W	

5 Custom Service

A&D proprietary service defines command operation to handle device setting and status.

A&D Service Characteristic allocates Size, Type, Command and Value to 20 bytes Readable/Writable area defined below.

Byte	0	1	2	3~19
Content	Size	Type	Command	Value

Size: Data length (bytes) from type to the end of Value

Type: 0; Read command. 1; Write command. 2; Response at Peripheral only

Profile Mapping

Handle	UUID Type	UUID	Attribute Type	Parameter Values or Description	Format	Endian	Permis sion	Len	Register	Value	Read/ Write	Comment
0x0020	16	0x2800	Primary Service Custom Service	Custom Service(0xF000)	UInt16		1	17		238BF0005A341B8D975C000D5690ABE4		
0x002D	16	0x2803	Characteristic Definition	Custom Characteristic Value	UInt8		1	1		NA		
0x002E	128	238BF0015A341B8D975C000D5690ABE4	Custom Characteristic	Set by Host(20octets max)	Ascii	Big	3	20	0x2830		R/W	

5.1 Blood Pressure Monitor (UA-651BLE)

Blood Pressure Monitor UA-651BLE

Command ID (0x00~0xFF)	Description	Operation		Data	Length	Data type (Binary or ASCII)	Format	description
		Connecte d	Before storing data					
0x01	Set Time	☐	☐	0Byte: Year Lower 2 digit 0D-FF 1Byte: Month 01-0C 2Byte: Day 01-1F 3Byte: hour 00-18 4Byte: minute 00-3B 5Byte: second 00-3B	6byte	Binary	ex) 2014/3/11 19:12:34 08 01 01 0E 03 0B 13 0C 22	Update date and time to device clock.
0x03	Disconnect	☐	☐	Non			02 01 03	Disconnect then move to stand-by mode
0x10	Unpair	☐	☐	Non			02 01 10	Unpair and move to stand-by mode.
0x12	Delete all memory	☐	☐	Non			02 01 12	Delete all data in buffer (Memory) and move to stand-by mode.
0xA6	Set Buffer size	☐	☐	0:NO buffer 1:30 data	1byte	Binary	ex) 30 data 03 01 A6 01	Delete all data in buffer and set buffer(memory) size
0xD6	Read Buffer size	☐	☐	Command only Response: refer to data part of setting device for production.	1byte	Binary	Request) 02 00 D6 Response) 03 02 D6 01	Read current buffer size (memory size) from A&D BLE device.